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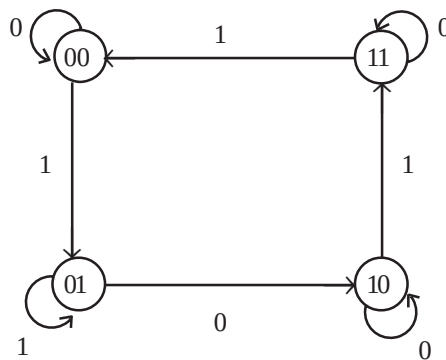
Roll No

EX-403 (GS)**B.E. IV Semester** Examination, June 2020**Grading System (GS)****Digital Electronics Logic Design - I****Time : Three Hours****Maximum Marks : 70****Note:** i) Attempt any five questions.

ii) All questions carry equal marks

1. a) i) Convert $(2F.C4)_{16} = (\quad)_8$
 ii) Convert $(762.013)_8 = (\quad)_{16}$
 b) Minimize the Boolean function
 $f(A, B, C) = \sum_m(0, 1, 3, 5) + \sum_d(2, 7)$
 using the mapping method is minimized SOP form.

2. For the state diagram shown in below figure, design a synchronous sequential circuit using JK flip flops.



3. A certain 12 bit BCD digital to analog converter has a full scale output of 9.99V.

- i) Determine the percentage of resolution
- ii) Determine converter's step size

4. a) Add the following numbers:

- i) $(7F)_{16}$ and $(BA)_{16}$
- ii) $(77)_8$ and $(107)_8$

- b) Differentiate between error-detecting and error-correcting code.

5. Write short notes on:

- a) SR latch with NAND and NOR gates
- b) Parity generator and checkers
- c) Alphanumeric code - ASCII code

OR

- a) Design a synchronous mod-6 counter.
- b) Design 4 bit ring counter.

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6. a) Explain the operation of JK flip-flop with neat diagram.
b) Explain the operation of master slave flip-flop.

OR

Design a 3-bit binary counter using TFF that has a repeated sequence of 6 states.
000-001-010-011-100-101-110. Give the state table, state diagram and logic diagram.

7. a) Minimize the Boolean function.
 $f(A, B, C, D) = \pi(0, 4, 5, 6, 7, 10, 14)$ using K-map method.
b) Using tabulation method simplify the Boolean function
 $F = \sum(0, 1, 2, 8, 10, 11, 14, 15)$

8. Write short notes on any two of the following:
a) PAL and PLA.
b) Parity generator and checker.
c) Full subtractor.
